

Solving One-Step Equations with Rational Coefficients

Grades 6–7 Expressions and Equations

Undoing a unit-fraction coefficient, multiplying by the reciprocal of a non-unit fraction, dividing to undo a decimal coefficient, working with a negative rational coefficient, and checking a solution by substitution

The one idea behind every problem. A one-step equation says “something was done to x , and the result is this.” You undo that one thing on *both* sides. When the coefficient is a fraction, the fastest undo is multiplying by its **reciprocal**; when it is a decimal, the undo is division.

Tape diagram convention used on this sheet



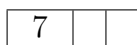
The whole bar is x . The bar is cut into equal parts, and the shaded or labelled parts are what the equation tells you about.

Reference diagram. No values shown — use the numbers given in each problem.

Directions.

- Show the undo step you used, written on both sides of the equation.
- Leave fraction answers as fractions in lowest terms unless the problem gives decimals.
- Write your final answer on the answer line.

1. A tape diagram is cut into 3 equal parts. One part is worth 7.



- (a) Write the equation the diagram shows, using x for the whole bar.
(b) Solve it. What is x ?

Answer: _____

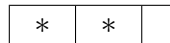
- 2.** A jar of marbles is being emptied. After a scoop, 0.4 of the marbles that were in the jar are on the table, and that is 6 marbles.

$$0.4x = 6$$

- (a) What single operation undoes multiplying by 0.4?
(b) Do it to both sides and solve for x .

Answer: _____

- 3.** A tape diagram is cut into 3 equal parts, and *two* of those parts together are worth 8.



the two starred parts together are worth 8

- (a) Explain in one sentence why x cannot be 4.
(b) Write and solve the equation $\frac{2}{3}x = 8$.

Answer: _____

4. A recipe uses one quarter of a bag of flour, which is 9 ounces.

$$\frac{x}{4} = 9$$

- (a) Sketch a bar cut into 4 equal parts and label one of them.
(b) Solve for x , the number of ounces in the whole bag.

Answer: _____

5. Mia claims that $x = 20$ is the solution of $\frac{3}{5}x = 12$.

Do not re-solve the equation. *Test* her answer instead: substitute 20 for x and evaluate the left side.

- (a) What value does the left side give?
(b) Is Mia right? Say in one sentence how the substitution proves it.

Answer: _____

- 6.** Solve the equation, where the coefficient is negative.

$$-\frac{3}{5}x = 12$$

- (a) Before solving, decide whether x is positive or negative, and say why.
(b) Multiply both sides by the reciprocal — sign included — and solve.

Answer: _____

- 7.** A phone plan charges 1.25 dollars per gigabyte, and this month's data charge was 7.5 dollars.

$$1.25x = 7.5$$

- (a) Will x be larger or smaller than 7.5? Say why in one sentence before you compute.
(b) Solve for x , the number of gigabytes used.

Answer: _____

8. Both sides of this equation are fractions.

$$\frac{5}{6}x = \frac{2}{3}$$

- (a) Write the reciprocal of the coefficient.
(b) Multiply both sides by it and simplify. Leave your answer as a fraction in lowest terms.

Answer: _____

9. This equation mixes a fraction coefficient with a decimal constant.

$$\frac{3}{8}x = 1.5$$

- (a) Rewrite 1.5 as a fraction.
(b) Solve by multiplying both sides by the reciprocal of the coefficient. Give your answer as a whole number.

Answer: _____

10. Challenge — find and fix the mistake. Devon solved $\frac{2}{3}x = 18$ like this:

$$\frac{2}{3}x = 18, \text{ so } x = \frac{2}{3} \cdot 18 = 12.$$

- (a) Substitute Devon's answer, $x = 12$, into $\frac{2}{3}x$. What value do you get?
- (b) Name the mistake in one sentence.
- (c) Solve the equation correctly. Write both your part (a) value and your corrected solution on the answer line.

Answer: _____